BearCart E-Commerce Analytics

Presented by
Data Dynamoz

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Introduction

This project analyzes BearCart's digital and transactional data to understand **customer behavior, marketing effectiveness, revenue performance, and refund impact.**

Using real-world datasets and Power BI dashboards, we transformed fragmented data into **clear, business-ready insights** that support smarter decision-making.

BUSINESS CONTEXT

Why This Analysis Matters :

BearCart is a growing e-commerce company that collects large volumes of data from:

- Website traffic
- Marketing campaigns
- Orders and products
- Refunds and user behavior.

However, due to **missing values, inconsistent formats, and data spread across multiple tables**, meaningful insights were difficult to extract.
This analysis bridges the gap between **data collection and business decisions**.

PROBLEM STATEMENT

Despite having rich digital and transactional data, BearCart lacked:

- Clear visibility into **which marketing efforts actually drive revenue**
- Understanding of **conversion behavior across sessions and campaigns**
- Insights into **product profitability and refund patterns**
- A unified view of the **end-to-end customer journey**

The challenge was to **clean, connect, and analyze** these datasets to generate **reliable and actionable insights**.

Dataset & scopes

Datasets Used & Scope of Analysis :

We analyzed 7 interconnected datasets, covering the complete funnel:

□ Website Sessions – traffic sources, devices, repeat behavior

□ Orders & Order Items – revenue, cost, profit

□ Products – product-level performance

□ Page Views – navigation and engagement behavior

□ Refunds – post-purchase dissatisfaction signals

Together, these datasets enabled analysis from
Session → Order → Revenue → Refund

DATA CLEANING & PROCESSING METHODOLOGY

How We Prepared the Data :

Before analysis, extensive preprocessing was performed:

- Handled missing UTM parameters and HTTP referrers as **valid organic/direct traffic**
- Standardized text values and corrected data types
- Removed duplicate records to prevent double counting
- Built a clean **relational data model** using primary–foreign keys
- Created derived metrics such as **Profit, AOV, Conversion Rate, and time-based features**
- Avoided unnecessary row deletion to preserve **real-world data integrity**

KEY PERFORMANCE INDICATORS (KPIs)

KPIs Used for Analysis

To translate data into business understanding, we focused on:

- **Total Sessions**
- **Total Orders**
- **Conversion Rate**
- **Total Revenue & Net Revenue**
- **Average Order Value (AOV)**
- **Revenue by Campaign, Source, and Product**
- **Refund Count & Refund Impact**

These KPIs were visualized dynamically using **Power BI dashboards**.

KEY INSIGHTS & FINDINGSKEY INSIGHTS & FINDINGS

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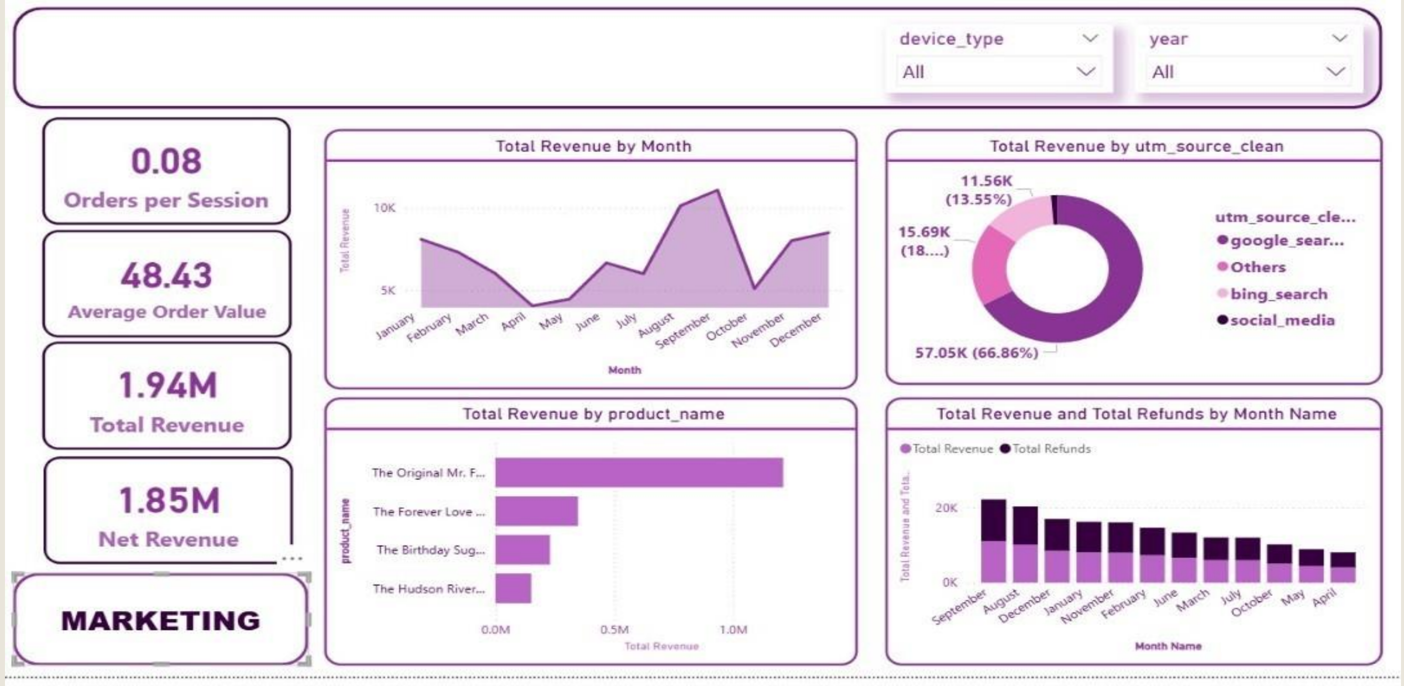
Insights Derived from the Analysis :

- A large share of traffic comes from organic and direct sources, indicating strong brand awareness
- Sessions linked to completed orders show higher engagement and purchase intent
- Google Search is the strongest contributor to both traffic and revenue
- Revenue is concentrated in a few products, following a Pareto (80/20) pattern
- Some campaigns generate traffic but fail to convert efficiently
- Refunds are concentrated in specific products, impacting profitability and trust

Revenue Dashboard – What's Working



Key Measures :

- **Total Revenue:** ~1.94M
- **Net Revenue:** ~1.85M
- **Average Order Value (AOV):** ~48.4
- **Orders per Session:** ~0.08

What the Visuals Show

- **Monthly Trend:**
- Revenue is seasonal, with clear peak months driving most sales
- **Product Performance:** A few products generate the majorof revenue(80/20 rule)
- **Channel Contribution:** Google Search is the strongest revenue source
- **Refund Analysis:** Refunds are controlled but spike in specific months

Key Point :

Revenue is strong but concentrated — growth depends on scaling top products, peak seasons, and minimizing refund losses.

Marketing Dashboard – What's Driving Growth

Key Measures

- **Total Sessions:** ~473K
- **Total Orders:** ~32K
- **Conversion Rate:** ~0.08
- **Campaign Revenue:** ~1.94M

What the Visuals Show :

- **Campaign Performance:**
- Non-brand campaigns drive the highest orders
- **Source Impact:** Google Search leads across campaigns and devices
- **Device Behavior:** Desktop users convert better than mobile
- **Campaign Distribution:** Few campaigns contribute most conversions

Final Business Recommendations

1. Improve Conversion Rates :
Encourage user sign-ups, optimize the mobile shopping experience, and ensure landing pages clearly match campaign intent.

2. Spend Smarter on Marketing :
Prioritize high-conversion channels, reduce spend on low-performing campaigns, and maintain consistent UTM tagging for accurate tracking.

3. Drive Profitable Growth :
Focus on high-margin products, introduce product bundling and upselling, and review pricing for low-margin items.

4. Reduce Refunds & Build Trust :
Improve product descriptions, strengthen quality checks, and optimize fulfillment to reduce dissatisfaction and returns.

Conclusion :
By converting raw, scattered data into clear insights and interactive dashboards, this project empowers BearCart to optimize marketing spend, improve conversions and data-driven growth.

Thank you

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